Guang-Xing Li, Ph.D.

Computational Physicist | Scientific Machine Learning | Multiscale Systems

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Summary

Computational physicist developing methods to extract structure, dynamics, and physical understanding from complex scientific data. For 15+ years, his work has progressed from physics-based structural diagnostics and multiscale representations to ScaleAware-JEPA, advancing self-supervised representation learning toward **ML-assisted knowledge discovery directly from scientific fields**.

Research scale: 111 scientific papers · 1,800+ citations · h-index 24 · €280k+ PI-led competitive funding · international research leadership · GPU/HPC scientific computing

Research Highlights — AI-Assisted Discovery & Multiscale Methods

ScaleAware-JEPA — *Representation learning for scientific knowledge discovery*

2026 · Sole author

A framework for using learned representations **not merely to predict predefined targets, but to expose scientifically meaningful structure before the objects of interest are specified**.

- Moves scientific ML **beyond predefined prediction toward knowledge discovery**: learns dense latent coordinates in which coherent structures can emerge **before objects, labels, or segmentation rules are specified**.
- Introduces **scale-aware latent prediction for continuous physical fields**, tying the JEPA prediction geometry to the intrinsic multiscale hierarchy of the data **rather than arbitrary fixed image patches**.
- Demonstrates a **domain-general discovery framework** across MHD turbulence, interstellar molecular gas, and urban structure; in astrophysical data, latent morphology recovers **physically distinct velocity–size relations**.
- Built the end-to-end research system, spanning multiscale representation, self-supervised GPU training, dense latent-field inference, and representation diagnostics.

Adjacent Correlation Analysis (ACA) — *Hidden regularities in complex spatial data*

2025 · Sole author

- Introduced a spatially aware correlation framework that **recovers local relationships obscured by global statistics**, transforming complex spatial patterns into interpretable vector fields in phase space.
- Connects data-driven regularity discovery with dynamical-systems structure: the recovered correlation vectors can represent the **vector field on an attracting manifold and expose distinct physical regimes**.

Constrained Diffusion Decomposition (CDD) — *Multiscale representation of continuous fields*

2022 · Sole author

- Developed a nonlinear-diffusion decomposition that assigns **pixel-registered physical scales** while avoiding scale mixing and filtering artifacts.
- Provides the **explicit multiscale coordinate** later used in learned physical representations.

G-virial — *Physics-based identification of gravitational structure*

2014–2018 · DFG-funded

- Developed an analytical–numerical framework for identifying **gravitationally significant structure** in turbulent media by connecting virial theory with simulated observables.

Research Leadership

Principal Investigator & Associate Professor **2018–Present**

Yunnan University (SWIFAR), China

- **Leading the transition of a multiscale-physics research program toward self-supervised learning and ML-assisted scientific knowledge discovery**, building on long-standing work in turbulence, structure formation, and scientific computing.
- **Built and directs the computational research environment** spanning large-scale simulations, heterogeneous scientific-data pipelines, HPC/Linux infrastructure, and GPU training.
- **Led a €60k Max Planck Partner Group** and **supervised PhD/MSc researchers** through scientific publications and international research placements.

Earlier Research Experience

DFG Research Fellow **2014–2018**
LMU Munich (Universitäts-Sternwarte), Germany

- Awarded a €220,000 DFG research fellowship to independently lead a multiscale structural-analysis program extracting physical understanding from observational and simulation data in large-scale Linux-based computing environments.
- Developed the G-virial method, combining analytical theory with numerical simulations to decode structural thresholds in highly turbulent media.

Ph.D. Researcher **2010–2014**
Max Planck Institute for Radio Astronomy, Germany

- Developed robust, scalable algorithms and computational models for multiscale structures in high-dimensional, heterogeneous environments; graduated *magna cum laude*.

Selected Scientific Discoveries

A kiloparsec-scale, long-lived superbubble across the Perseus arm — *Nature Communications* (2025)
Identified a large, long-lived, slowly expanding Galactic superbubble; **co-corresponding author.**
B. Chen, **G.-X. Li**, H. Yuan, *et al.*

Cosmic-ray shielding by dense molecular gas — *Nature Astronomy* (2023)
Demonstrated effective suppression of $\lesssim 10\,\mathrm{GeV}$ cosmic rays inside dense molecular clumps.
R.-Z. Yang, **G.-X. Li**, E. de Oña Wilhelmi, *et al.*

Flyby-induced spirals in a massive protostellar disk — *Nature Astronomy* (2022)
Interpreted spiral structure in a massive Keplerian disk as the dynamical signature of a stellar flyby; **co-corresponding author; led interpretation and analytical modeling.**
X. Lu, **G.-X. Li**, Q. Zhang & Y. Lin.

Coherent velocity wave in the Radcliffe Wave — *MNRAS Letters* (2022)
Discovered a coherent wave-like velocity pattern across the Galactic-scale Radcliffe structure; **first author.**
G.-X. Li & B.-Q. Chen.

Education

Ph.D. in Astrophysics and Computational Modeling **2014**
Max Planck Institute for Radio Astronomy & University of Bonn, Germany *(magna cum laude)*
M.Sc. in Astrophysics **2010**
University of Science and Technology of China (USTC)
B.Sc. in Electronics **2007**
Peking University (PKU)

Technical Skills

Methods: Self-supervised learning · representation learning · PyTorch · GPU/HPC computing · multiscale analysis · hydrodynamics/MHD